BIOL-1 Practice Test 04

Exam 03 Preparation (Modules 12–16)

Instructions: This practice test covers material from **Module 12** (Darwin and Evolution) through **Module 16** (Capstone Systems Synthesis). Answer all questions to the best of your ability to prepare for Exam 03.

✂️ TEAR-OFF ANSWER SHEET **Write your name and date above. Turn in this page when instructed.**
Multiple-choice answers 1. ______ 2. ______ 3. ______ 4.
______ 5. ______ 6. ______ 7. ______ 8. ______ 9. ______ 10.
______ 11. ______ 12. ______ 13. ______ 14. ______ 15.
______ 16. ______ 17. ______ 18. ______ 19. ______ 20.
______ 21. ______ 22. ______ 23. ______ 24. ______ 25.
______ ## Fill-in answers 1.
________________________________ 2.
________________________________ 3.
________________________________ 4.
________________________________ 5.
________________________________ 6.
________________________________ 7.
________________________________ 8.
________________________________ 9.
________________________________ 10.
____
One answer per line; use the question numbers printed in the booklet.

QUESTION BOOKLET — FREE RESPONSE **Answer the free-response questions on the lined space provided.** ## Part C: Short Answer (4 questions) **36.** **(Module 12)** In two or three sentences, define **evolutionary fitness** as used in this course, and name **one** line of **evidence** for common descent (fossils, comparative anatomy, biogeography, or molecular) with a brief what-it-shows.

37. **(Module 13)** Distinguish **bottleneck** and **founder** effects. Then name the three **modes of natural selection** on a continuous trait (directional, stabilizing, disruptive) in one line each.

38. **(Module 14)** Give one **prezygotic** and one **postzygotic** example of a **reproductive barrier** (one sentence each). In one sentence, contrast **allopatric** and **sympatric** speciation.

39. **(Module 15)** Contrast **density-dependent** and **density-independent** limiting factors with one **example of each** (one sentence on each type). In two sentences, use the **~10%** energy rule to explain **why** food chains with many links support **fewer** apex consumers than a short chain from the same primary production.

40. **(Module 16)** Build a short systems explanation: choose one biological scenario from this unit and connect **one evolutionary idea**, **one population/ecology idea**, and **one piece of evidence** into a single cause-and-effect explanation.

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QUESTION BOOKLET — MULTIPLE CHOICE AND FILL-IN **Question booklet: answer the questions below using the tear-off sheet for Parts A and B.** ## Part A: Multiple Choice (25 questions) *Choose the best answer for each question.* **Module 12: Darwin and Evolution** **1.** In the logic of natural selection, which observation states that more offspring are produced than can usually survive to reproduce? A) Inheritance of acquired characters B) Differential reproductive success C) Overproduction of offspring D) Stabilizing selection only **2.** A trait increases **evolutionary fitness** in a given environment if individuals with that trait tend to: A) Contribute more alleles to the next generation than competitors do B) Acclimate in one lifetime to match the parent phenotype C) Look larger than their competitors D) Survive long enough to be counted, even with zero offspring **3.** Which statement is **most** consistent with **descent with modification**? A) Each species was created in its final form and does not change B) Lineages change over time; related species share ancestors with modification C) Only fossils evolve, not living species D) The strongest individual in a species can change the population's heritable gene pool in one lifetime, without reproduction **4.** Homologous forelimb bones in a human, a cat, and a whale are used as evidence for evolution because they: A) Suggest a shared ancestry with structural change for different uses B) Disprove the existence of a fossil record C) Prove that whales evolved from modern humans D) Show identical functions in all three species **5.** A population of moths in a clean forest is mostly light-colored; in a sooty industrial region the population becomes mostly dark. The clearest **selection** label for that shift, if the environment favors the dark extreme, is: A) Stabilizing selection B) The founder effect C) Directional selection D) Disruptive selection **6.** Which option best states why **populations** evolve, not **individuals**? A) One hungry bird cannot affect allele frequencies at all (bird effects are not evolution) B) Natural selection has nothing to do with heritable variation C) Evolution is a change in allele **frequencies in a population** across generations D) Individuals can change their DNA by lifting weights **Module 13: How Populations Evolve** **7.** The definition of **microevolution** used in this course is: A) A **change in allele frequencies** in a population over time B) A change in an individual's size during one lifetime C) A synonym for any mutation in a body cell only D) The origin of a new phylum in one generation **8.** A few finches from the mainland start a new island population. The island gene pool reflects only the alleles the founders brought. This is: A) Stabilizing selection on beak size B) The bottleneck effect C) The **founder** effect (genetic drift) D) Convergent evolution **9.** A volcanic eruption kills 95% of a squirrel population

at random, then survivors repopulate. The loss of many alleles from the pre-eruption pool is: A) A founder effect in a new island colonization from another continent only B) A **bottleneck** (genetic drift) C) A proof that survivors were the “most fit” for every possible future environment D) Interspecific competition in the absence of intraspecific variation **10.** In **stabilizing** selection on a trait (for example, birth weight in some human data sets): A) Allele frequencies must stay exactly at Hardy–Weinberg equilibrium forever B) The population’s average always moves to one tail of the range C) The **intermediate** phenotype is favored; extremes are selected against D) Both extreme phenotypes are favored **11.** The Hardy–Weinberg equilibrium model is useful mainly because it: A) Describes a real population that never evolves B) Serves as a **mathematical baseline** to measure evolutionary change in real data C) Guarantees $p^2 + 2pq + q^2 = 1$ in every generation without assumptions D) Replaces the biological species concept **12.** The movement of alleles into or out of a population through **migration and mating** is: A) Genetic drift in an infinitely large population B) **Gene flow** C) A postzygotic barrier to reproduction D) Non-random mating only **Module 14: Macroevolution** **13.** Under the **biological species concept**, two groups are different species if they: A) Are **reproductively isolated** in nature (they do not interbreed to produce viable, fertile offspring **between** the two groups, while within each group interbreeding can occur) B) Have different number of cells C) Live on different trees in the same forest, without any other test D) Look different in a field guide, regardless of interbreeding **14.** Fireflies of two species use different **flash** patterns, so they rarely attempt mating. This isolating mechanism is best classified as: A) The bottleneck effect B) Postzygotic reduced hybrid fertility C) A mechanical postzygotic barrier D) A **prezygotic** **behavioral** barrier **15.** A mule, the hybrid of a horse and a donkey, is typically sterile. This is an example of: A) A prezygotic gametic barrier B) The founder effect C) A **postzygotic** barrier to gene flow (reduced hybrid fertility) D) Stabilizing selection in horses **16.** A new mountain range **physically splits** a population; over long time, the two lineages no longer interbreed. The classic geographic mode of speciation is: A) A requirement that drift never occurs B) **Allopatric** speciation C) Sympatric speciation in one meadow D) The bottleneck effect in both lineages on purpose **17.** A plant can form a new, reproductively isolated **polyploid** lineage in a few generations, often in the same area as a parent type. This route is a frequent example of: A) The bottleneck effect only B) Only macroevolution in animals, not plants C) A prezygotic postzygotic hybrid D) **Sympatric** speciation (e.g. via polyploidy) **18.** The statement “macroevolution is different chemistry from microevolution” is **false** in this course because: A) The fossil record is irrelevant to modern biology B) All macroevolution happens in a single meiosis C) No DNA is involved in speciation D) The same **population**-level processes (selection, drift, flow, etc.) can accumulate to reproductive isolation and larger-scale diversity **Module 15: Population and Systems Ecology** **19.** A population introduced to a new habitat with abundant resources for a time shows growth that **accelerates** (as long as the ceiling is not yet reached). That early phase is classic for: A) An **exponential (J**-shaped) phase of growth B) A carrying capacity of zero by definition C) A flat line at carrying capacity from year one only D) Only density-independent regulation, never K **20.** A population levels off as resources become scarce, approaching a relatively stable size. That ceiling is: A) A genetic bottleneck B) **Carrying capacity (K)** in a logistic (S) model C) A proof that 100% of energy moves to the next level D) A synonym for the founder effect **21.** A fungal disease that spreads more easily in **dense** host stands is an example of a factor often: A) Only density-independent like wind B) Unrelated to per-capita rates C) Never biological D) **Density-dependent** in its impact on the population **22.** A severe freeze kills a **similar fraction** of a plant population across sparse and dense patches (the freeze does not “care” about crowding the way a contagion might). Treated as a **limiting** event in textbook classification, the freeze is **most** like: A) A **density-independent** disturbance (still harmful to the population) B) A biogeochemical cycle for nitrogen only C) Only density-dependent competition for light D) A trophic level **23.** The **~10% rule** between trophic levels implies that, compared to eating **producers** directly, a diet of **higher** trophic levels (many steps) to deliver the same human calories will tend to need: A) Less primary production because carnivores are 100% efficient B) The same amount of energy at every level, by definition of food chains C) Decomposers to add energy from the sun to dead matter D) **More** total primary production at the base of the food web **24.** In an ecosystem, **chemical elements** (e.g. carbon, nitrogen in nutrients) are **recycled**, while the energy fixed by **photosynthesis** is eventually: A) **Lost** largely as **heat** along the way; not recycled the way a carbon atom can be B) Transferred 100% from producer to all consumers combined C) Only stored in rocks, not in life D) Re-used indefinitely by the same photon inside every cell with no loss **25.** Fungi and bacteria in soil return nutrients from dead material to **forms** plants can use again. In contrast to energy, this recycling pattern supports the idea that: A) Decomposers are always apex predators in every web B) Energy cycles exactly like water in the stratosphere only C)

Producers do not use soil nutrients D) Decomposers are central to **nutrient** cycling for producers, while **energy** flows in one direction through the system --- ## Part B: Fill in the Blank (10 questions) **Word Bank:** Allopatric, allele frequencies, decomposers, density-dependent, gene flow, homologous, K, prezygotic, stabilizing selection, sympatric **26.** A population’s long-term, stable ceiling in a logistic (S) model is the carrying capacity, symbol ________________. **27.** The formal focus of **microevolution** is a change in ____________________ over generations in a population. **28.** A barrier that works **before** a zygote—such as different breeding times—is a ____________________ barrier to reproduction (category name). **29.** A limiting factor (such as a contagious disease) whose impact on survival or reproduction can intensify with crowding is often ____________________. **30.** The movement of alleles into or out of a population through migration and mating is called ____________________. **31.** The forelimb bone pattern shared by many tetrapods with different functions (human arm, bat wing, whale flipper) is ____________________ structure (in comparative anatomy, from common ancestry with modification). **32.** Fungi and many bacteria, which break down dead material and return elements to the soil, are ____________________ in nutrient cycling (category name). **33.** Modes of natural selection: when the **middle** of a trait’s range is favored and the extremes are selected against, the pattern is often ____________________. **34.** A river splits a continuous population; the two sides evolve in isolation. That geographic mode of speciation is often ____________________ speciation. **35.** A new species arises **without** geographic separation, as in some plant **polyploidy** lineages, is a textbook example of ____________________ speciation. ---